

Brainstorming & Idea Prioritization

Project Title: Rising Water – Flood Prediction System using Machine Learning

1. Introduction

Floods are among the most destructive natural disasters. The Rising Water project uses Machine Learning to predict flood probability based on environmental factors and provides results through a Flask web application.

2. Brainstorming Process

S.No	Idea	Advantages	Limitations
1	Weather Forecast	Predicts weather	No flood prediction
2	Rainfall Prediction	Useful for agriculture	Limited flood analysis
3	Water Quality Monitoring	Environmental monitoring	Not flood focused
4	Flood Prediction System	Early warning & disaster support	Needs historical data

3. Objectives

- Develop an ML model to predict flood probability.
- Build a Flask web application.
- Provide early flood risk estimation.
- Improve disaster preparedness.

4. Technologies

Technology	Purpose
Python	Programming
Pandas	Data Processing
NumPy	Numerical Computing
Scikit-learn	Machine Learning
XGBoost	Prediction Model
Flask	Web Application
HTML/CSS	Frontend
Joblib	Model Storage

5. Conclusion

After evaluating multiple ideas, the Flood Prediction System was selected because it provides significant social impact and demonstrates practical use of Machine Learning.